Workflow Refinery

A human-guided compute constellation – concept, not deployed system

Dots Workflow Refinery Demo

October 2026

The proposed second layer

Turn explicitly selected repeated agent workflows into evaluated, versioned program candidates.

Hypothesis

A narrower DSPy program on a smaller model may reduce compute cost while maintaining or improving task quality. This must be tested, not assumed.

The thesis needs boundaries

- The claim that all work can be reduced to agent-controlled compute is a research thesis, not established fact.
- Human judgment, relationships, physical work, legal duties, and consent still matter.
- No workflow or conversation enters the proposed pipeline without explicit scope and review.

A controlled refinement loop

1. Owner selects and authorizes one bounded task.
2. Reviewers redact examples and adjudicate a golden dataset.
3. Compare DSPy and smaller-model candidates on held-out cases.
4. Human approves a versioned artifact bundle.
5. Monitor the approved scope and roll back on failure.

What the golden dataset contains

Versioned evidence

Representative inputs, expected outcomes, provenance, labeling rules, and edge cases.

Evaluation discipline

A held-out split, disagreement adjudication, error categories, and regression checks.

Refinement means improving labels and coverage under review, not silently collecting more private material.

Model selection is a measured decision

Compare baseline and candidate on the same cases. Record model identity, program version, dataset version, quality, latency, and total inference cost.

Promotion gate

A lower-cost candidate advances only when it meets predefined quality and safety thresholds and a human reviewer approves.

Possible domains, no live actions

- Internal: support triage, document tagging, task routing.
- Future research: job discovery, contract research, consulting lead qualification.
- This public demo performs no ingestion, applications, bids, outreach, contract closing, training, or provider calls.

What exists today

- React and Storybook illustrate selection, review, simulated evaluation, promotion, and rollback.
- Playwright tests the UI gates and records a synthetic walkthrough.
- All sample records and scores are invented; 78% and 92% are simulated interface values.

Public source: github.com/newdarwindev/dots-workflow-refinery-demo